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# The Memory Myth: A Reconstruction Paradigm for Emergency Command Debriefing

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## Abstract

Post-incident debriefing in emergency services depends overwhelmingly on verbal recall — a practice we term the *memory myth*. Research on stress and memory demonstrates that acute physiological arousal, ubiquitous during emergency command, selectively degrades the episodic memory needed to reconstruct what actually happened. The critical moments — the first four minutes of a briefing, the command handoff, the moment a GRIP protocol step was missed — are precisely those that are least reliably encoded. We propose an alternative: the *reconstruction paradigm*, which recovers event structure from multimodal data rather than recall. Reconstruction is explicitly partial, multi-source, and discovery-oriented; it makes no claim to comprehensive measurement. We present **SMASHFIRE**, a platform implementing this paradigm for Brandweer Rotterdam-Rijnmond through eleven analysis modules covering speech transcription, speaker attribution, communication patterns, stress-consistent speech and movement signatures, drone-based situational awareness, and GRIP-1 compliance detection. We argue that framing the system as a *map* rather than a measurement device is both epistemologically honest and practically consequential: it sets appropriate expectations, invites collaborative interpretation, and grounds post-incident learning in recoverable evidence rather than contested memory.

**Keywords:** emergency debriefing, multimodal analysis, reconstruction paradigm, post-incident learning, GRIP protocol, crisis informatics

## 1. Introduction

When a GRIP-1 emergency ends and the incident commander gathers the team for debriefing, a question is implicitly assumed to be answerable: *What happened?* This paper argues that the assumption is wrong, and that its wrongness has systematic consequences for how emergency services learn from incidents.

Post-incident debriefing in fire and rescue services typically proceeds through three channels: verbal recall from participants, manual notes taken under operational pressure, and the commander's reconstruction of the timeline. Each channel is subject to well-documented biases. Recall degrades under acute stress (Roozendaal et al., 2009). Notes are selective and retrospective. And the timeline — the authoritative account of *who did what when* — is composed after the fact, drawing on the same compromised memories. The product of this process is not a record of what happened. It is a consensus reconstruction of what participants remember happening, presented as though the two were equivalent.

We call this conflation the *memory myth*. The myth is not that memory is involved in debriefing — of course it is. The myth is that memory-based debriefing reliably recovers the event structure needed for post-incident learning. The literature on stress and memory provides a coherent account of why this reliability fails precisely where it is needed most: under the conditions of acute arousal that characterise emergency command, episodic memory encoding is selectively impaired (Deffenbacher et al., 2004; Starcke & Brand, 2012). Firefighters emerging from a GRIP-1 briefing carry elevated cortisol and norepinephrine, narrowed attentional focus, and reduced working-memory capacity. Their recall of the briefing — the who, the when, the sequence of confirmations and missed steps — is accordingly unreliable.

Against this background, we propose the *reconstruction paradigm*: the systematic recovery of event structure from multimodal data rather than from recall. The paradigm has three defining properties. It is *multi-source*: speech transcripts, speaker attribution, body movement, positional data, and aerial imagery each provide a partial view, and no single channel is sufficient. It is *incomplete by design*: reconstruction does not claim to recover everything that happened; it is explicit about what it does and does not reach. And it is *discovery-oriented*: reconstruction routinely surfaces patterns that participants would not have reported — a silence that lasted eight seconds, a step mentioned once but never confirmed, a speaker who dropped from the conversation at a critical moment — because these patterns were not consciously registered under load.

We also articulate a deliberate distinction between reconstruction and *measurement*. Measurement implies a known quantity, a validated scale, and a precision claim. We are not in a position to *measure* stress in an emergency briefing in any clinically robust sense. What we can do is identify patterns in speech acoustics and movement kinematics that are consistent with elevated physiological arousal — patterns that stress researchers recognise as proxies (Hellhammer et al., 2009). This is reconstruction: building a picture from mutually consistent fragments, with explicit epistemic humility about completeness. The distinction is not semantic. It determines whether the system is positioned as a diagnostic instrument (measurement) or a collaborative evidence base (reconstruction / map), and the two positions have very different implications for how practitioners use the output.

We present **SMASHFIRE** — the emergency-services extension of the SMASH (Synthesis and Multimodal Analytics System for Humanities) platform (Owoyele et al., 2026) — as a concrete implementation of the reconstruction paradigm, developed in partnership with Brandweer Rotterdam-Rijnmond under the NWO/ZonMw Impact Explorer 2025 programme. SMASHFIRE operates through eleven analysis modules and produces a structured map of a briefing: who spoke when, what was said, which GRIP-1 steps were present or absent, and what arousal-consistent patterns were detectable.

The remainder of this paper is organised as follows. Section 2 develops the memory myth argument. Section 3 formalises the reconstruction paradigm. Section 4 presents the SMASHFIRE architecture. Section 5 describes the eleven reconstruction modules. Section 6 demonstrates the system on a simulated incident briefing. Section 7 discusses implications and limitations. Section 8 situates the work in related research. Section 9 concludes.

## 2. The Memory Myth

### 2.1 What the myth claims

The memory myth rests on a tacit syllogism. First premise: participants in an emergency briefing were present and attentive. Second premise: present and attentive witnesses remember what they witnessed. Conclusion: debriefing recovers what happened.

Each premise is partially true. Participants were present. Memory did encode something. But the specific kind of encoding required for post-incident learning — temporally ordered, agent-attributed, causally connected episodic memory (Tulving, 1983) — is the kind most vulnerable to the conditions of emergency command.

### 2.2 Stress and episodic memory

Acute stress activates the hypothalamic-pituitary-adrenal (HPA) axis and the sympathetic-adrenomedullary system, releasing cortisol and catecholamines into the bloodstream within seconds to minutes of a stressor. These hormones have well-documented effects on memory. Cortisol impairs prefrontal cortical function and working memory capacity (Arnsten, 2009), reducing the available cognitive resources for encoding contextual details. Norepinephrine acts on the amygdala and hippocampus in opposing ways: it enhances the encoding of emotionally salient content while impairing the encoding of peripheral contextual detail (Roozendaal et al., 2009). The net effect is a systematic dissociation: the emotional salience of an event is well-remembered, but the *sequence*, the *attribution*, and the *peripheral detail* that constitute an accurate timeline are not.

This dissociation has a cruel irony when applied to emergency debriefing. Commanders can recall *that* the briefing felt chaotic. They cannot reliably recall *when* the chaos began, *who* introduced the confusion, or *which specific step* was the inflection point. Precisely the information needed for post-incident learning is the information most degraded by the stress that accompanies the event.

### 2.3 The peak-end effect and narrative reconstruction

A second mechanism compounds the problem. Kahneman et al.'s 1993 peak-end rule describes how people evaluate experienced events: memory is dominated by the most intense moment (the peak) and the final moment (the end), not by a temporal integration of the whole. In emergency command, the peak is typically the most stressful tactical decision; the end is the resolution of the incident. Everything between arrival and resolution is subject to substantial compression and reconstruction in memory.

What participants report in debriefing is therefore not a record but a *narrative*: a causally coherent account constructed post-hoc from salient fragments (G. A. Klein, 1998). This is adaptive — narrative coherence aids learning — but it is not the same as temporal accuracy. The narrative elides the eight-second silence when no one confirmed the perslucht check. It glosses the moment the commander's speech rate doubled. It smooths the transition that felt seamless but in footage lasted twenty-three seconds of overlapping talk.